☐ Nuclear model: Degenerate Fermi Gas model

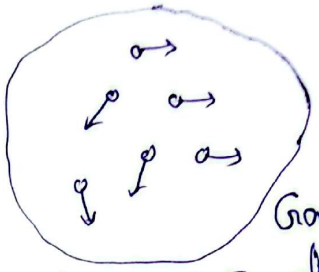
One Body Description ⟶ Many Body Description

mass: m → position $x(t)$

velocity: $v(t)$

momentum $p = mv$

Kinetic energy $E_K = \frac{1}{2}mv^2 = \frac{p^2}{2m}$



Charle's and Boyle's law

Pressure: P, Volume: V

Temperature: T

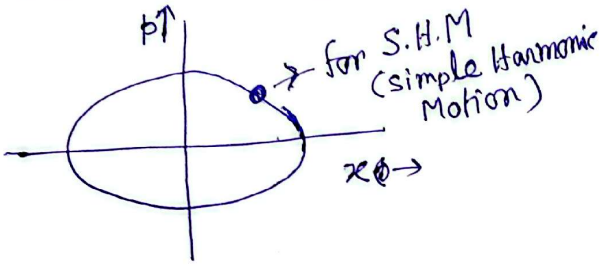
$\frac{PV}{T}$ = constant = R (Gas const.)

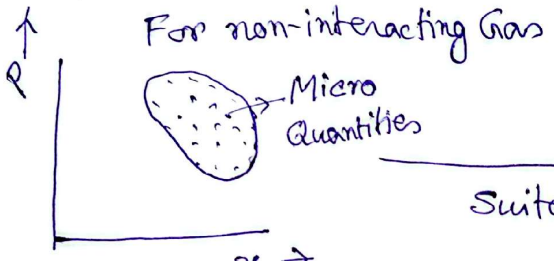
$= N_A K_B$

Avogadro No. $\sim 6 \times 10^{23}$

Boltzmann constant 1.3×10^{-23} J/K

Phase-space diagram





Suitable summation → Macro quantities (P, N, U etc.)

imagine $3N_A$ x-axes and $3N_A$ p-axes

Statistical Mechanics

$\sum \equiv \int \frac{d^3x\, d^3p}{h^3} f(p)$

Thermal distribution function

$\sum 1 = N$

$\sum E_K = U$

$\sum \frac{pv}{3} = PV$

Boson → Spin is integer $\hbar$ $\Rightarrow S = (0, 1, 2, \cdots) \times \hbar$

Fermion → " " half-integer of $\hbar$

$\Rightarrow S = \left(\frac{1}{2}, \frac{3}{2}, \cdots\right) \times \hbar$

$$-PV = \Phi = -KT \ln Z_G$$

$$= -KT \sum \ln Z = -KT \sum \ln\left[1 + e^{-\beta(\epsilon-\mu)}\right] \quad \text{for Fermion}$$

$$= -KT \sum \ln\left[1 - e^{-\beta(\epsilon-\mu)}\right]^{-1} \quad \text{for Boson}$$

In general, we can write ..

$$-PV = \Phi = \frac{-KT}{\eta} \sum \ln\left[1 + \eta e^{-\beta(\epsilon-\mu)}\right] \quad \begin{cases} \eta = +1 \ (F) \\ \eta = -1 \ (B) \end{cases}$$

$$= \frac{-KT}{\eta} \int \frac{V d^3p}{h^3} \ln\left[1 + \eta e^{-\beta(\epsilon-\mu)}\right]$$

$$N = -\left(\frac{\partial \Phi}{\partial \mu}\right)_{T,V} = \frac{1}{\eta} \sum \frac{\eta e^{-\beta(\epsilon-\mu)}}{1 + \eta e^{-\beta(\epsilon-\mu)}}$$

$$= \sum \frac{1}{e^{\beta(\epsilon-\mu)} + \eta}$$

$$= \int \frac{V d^3p}{h^3} \frac{1}{e^{\beta(\epsilon-\mu)} + \eta}$$

$$\left.\begin{matrix} N_{FD} \\ N_{BE} \\ N_{MB} \end{matrix}\right\} = \int \frac{V d^3p}{h^3} f_0 \quad \begin{cases} \frac{1}{e^{\beta(\epsilon-\mu)} + 1} \\ \frac{1}{e^{\beta(\epsilon-\mu)} - 1} \\ e^{-\beta(\epsilon-\mu)} \end{cases}$$

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Classical distribution ⟶ Quantum distribution

$f_0 = e^{-\beta(\epsilon-\mu)}$ (MB)

$f_0 = \frac{1}{e^{\beta(\epsilon-\mu)}+1}$ (FD)

$f_0 = \frac{1}{e^{\beta(\epsilon-\mu)}-1}$ (BE)

$\sum f_0 =$

$$N = \frac{V}{h^3}\int d^3p\, f_0(\epsilon)$$

$$= \frac{V}{h^3}\int d^3p\, \frac{1}{e^{\beta(\epsilon-\mu)}+\eta}$$

$\eta = 0$ (MB)

$\eta = 1$ (FD)

$\eta = -1$ (BE)

$\sum \epsilon f_0 =$

$$U = \frac{V}{h^3}\int d^3p\,[\epsilon]\, f_0(\epsilon)$$

$$= \frac{V}{h^3}\int d^3p\,[\epsilon]\, \frac{1}{e^{\beta(\epsilon-\mu)}+\eta}$$

$$PV = -\Phi = \frac{KT}{\eta}\sum \ln\{1+\eta e^{\beta(\mu-\epsilon)}\}$$

$$PV = \frac{KT}{\eta}\int \frac{V d^3p}{h^3}\ln\{1+\eta e^{\beta(\mu-\epsilon)}\}$$

$$= \frac{V}{h^3}\int d^3p\,[\,?\,]\,\frac{1}{e^{\beta(\epsilon-\mu)}+\eta}$$

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check!

$$\ln Z = -\frac{1}{\eta}\sum_i \ln\left[1+\eta e^{\beta(\mu-\epsilon)}\right]$$

$$= \lim_{\eta\to 0} \frac{-\sum_i \frac{\partial}{\partial\eta}\ln\left[1+\eta e^{\beta(\mu-\epsilon)}\right]}{\frac{\partial\eta}{\partial\eta}}$$

$$= -\sum_i \lim_{\eta\to 0} \frac{e^{\beta(\mu-\epsilon)}}{1+\eta e^{\beta(\mu-\epsilon)}}$$

$$= -\sum_i e^{\beta(\mu-\epsilon)}$$

$$pV = \Phi = -KT\ln Z = KT\sum_i e^{\beta(\mu-\epsilon)} = KT\underbrace{\int \frac{d^3x d^3p}{h^3} e^{-\beta(\epsilon-\mu)}}_{N}$$

$$N = +\left(\frac{\partial\Phi}{\partial\mu}\right)_{T,V} = KT\times\beta\sum e^{-\beta(\epsilon-\mu)} = \int \frac{d^3x d^3p}{h^3} e^{-\beta(\epsilon-\mu)}$$

~~$S = \left(\frac{\partial\Phi}{\partial T}\right)$~~

P_{FD}

P_{BE}

P_{MB}

$$= \int KT \int \frac{V d^3p}{h^3}\left[\ \right]$$

$$-\ln\left\{1+e^{-\beta(\epsilon-\mu)}\right\}$$

$$\ln\left\{1-e^{-\beta(\epsilon-\mu)}\right\}$$

$$e^{-\beta(\epsilon-\mu)}$$

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$$n = g\int_0^{p_F} \frac{d^3p}{h^3}\cdot 1 = g\,\frac{4\pi}{h^3}\left(\frac{p_F^3}{3}\right)$$

$$E_F = \frac{p_F^2}{2m}$$

$$= \sqrt{p_F^2c^2 + m^2c^4}$$

$$\Rightarrow p_F = \left[\frac{3h^3}{4\pi g}\,n\right]^{1/3}$$

$$= g\,\frac{4\pi}{3h^3}(2mE_F)^{3/2} \quad \text{N.R.}$$

$$= g\,\frac{4\pi}{3h^3}\left(\frac{E_F^2}{c^2} - m^2c^2\right)^{3/2} \quad \text{R}$$

$$= g\,\frac{4\pi}{3h^3}\,\frac{E_F^3}{c^3} \quad \text{R for } m=0$$

$$\epsilon = g\int_0^{p_F} \frac{d^3p}{h^3}\,E = \frac{g}{h^3}\int_0^{p_F} 4\pi p^2 dp\left(\frac{p^2}{2m}\right)$$

$$= \frac{g}{h^3}\,\frac{4\pi}{2m}\,\frac{p_F^5}{5} \quad \text{NR}$$

$$= \frac{g}{5h^3}\,\frac{4\pi}{2m}\,(2mE_F)^{5/2}$$

$$\boxed{\langle E\rangle = \frac{\epsilon}{n} = \frac{3}{5}E_F}$$

$$P = g\int_0^{p_f} \frac{d^3p}{h^3}\left(\frac{pv}{3}\right) \qquad v = \frac{p}{m}$$

$$= \frac{g}{3h^3}\int_0^{p_F} 4\pi p^2\left(p\,\frac{p}{m}\right)$$

$$= \frac{g}{3h^3}\,\frac{4\pi}{m}\,\frac{p_F^5}{5} \Rightarrow p_F = \left[\frac{15Ph^3m}{g\,4\pi}\right]^{1/5}$$

$$= \frac{g\,4\pi}{15h^3m}\left[\frac{3h^3}{4\pi g}\,n\right]^{5/3}$$

$$= \underbrace{\frac{1}{5m}\left[\frac{3h^3}{4\pi g}\right]^{2/3}}_{K} n^{5/3}$$

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Hydrodynamical Equilibrium

$$\frac{1}{\rho_e}\frac{dP_e}{dr} = \frac{GM}{R^2}$$

$$M = \frac{4}{3}\pi R^3 \rho_e$$

$$\rho_e = \frac{M}{\frac{4}{3}\pi R^3} \propto \frac{M}{R^3}$$

$$\int_0^R \frac{dP_e}{dr}dr = \int_0^R \frac{GM}{R^2}\,\frac{M}{\frac{4}{3}\pi R^3}\,dr$$

$$P = \frac{GM^2}{\frac{4}{3}\pi}\,\frac{R^{-5+1}}{-5+1}$$

$$P_e \propto \frac{M^2}{R^4} \quad \text{——①}$$

But degenerate pressure $P_e \propto \rho_e^{5/3}$

$$\propto \frac{M^{5/3}}{R^5} \quad \text{——②}$$

$$\Rightarrow \frac{M^{1/3}}{R} \propto \frac{\rho_e}{P_e} \Rightarrow R \propto \frac{1}{M^{1/3}} \Longrightarrow \rho_e \propto \frac{M}{R^3}$$

$$\rho_e \propto M^2$$

For Relativistic

$$P = \frac{m_e c^2}{8\pi^2 \lambda_e^3}\,\phi(x_e) \qquad x_e = \frac{P_F}{m_e c}$$

$$\rho = \hat{\mu} m_p c^2 n_e = \cdot$$

$$\propto \begin{cases} P \propto \rho^{5/3} & \text{for } x_e \ll 1 \\ \propto \rho^{4/3} & \text{for } x_e \gg 1 \end{cases}$$

$$\left(\frac{9\pi M}{4\hat{\mu} m_p}\right)^{4/3} - \frac{3\pi\alpha}{\hbar c} GM^2 = \left(\frac{R}{\lambda_e}\right)^2 \left(\frac{9\pi M}{4\hat{\mu} m_p}\right)^{2/3}$$

>0 $\qquad$ >0

$$M \le \left(\frac{9\pi}{4\hat{\mu} m_p}\right)^2 \left(\frac{\hbar c}{3\pi\alpha G}\right)$$

$$m_p = \sqrt{\frac{\hbar c}{G}} = 10^{19}\ \text{GeV}/c^2$$

$\sim$ (some GeV) $\times 10^{19}$

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